

Find the area of triangles

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Find the area of a triangle with base 15 cm and perpendicular height 8 cm. Copy or complete the first step.

2. A triangle has area 54 cm² and base 12 cm. Find its perpendicular height.

3. A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.

4. Miro says: Miro uses the sloping side as the height and forgets to halve a triangle's area. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find the area of a triangle with base 15 cm and perpendicular height 8 cm. Copy or complete the first step.
Answer 60 cm².
2. A triangle has area 54 cm² and base 12 cm. Find its perpendicular height.
Answer 9 cm.
3. A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.
Answer 80 cm².
4. Miro says: Miro uses the sloping side as the height and forgets to halve a triangle's area. Is this correct? Explain.
Answer Mark the perpendicular height, use base times height, and divide by two only for a triangle.

Find the area of triangles

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Find the area of a triangle with base 15 cm and perpendicular height 8 cm.

6. Check this result using an inverse, estimate or known fact: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.

2. A triangle has area 54 cm² and base 12 cm. Find its perpendicular height.

7. Prove or explain your reasoning: A rectangle is 14 cm by 9 cm. Find its area and perimeter.

3. A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.

8. Identify and correct this misconception: Miro uses the sloping side as the height and forgets to halve a triangle's area.

4. Solve this independently and show every stage: Find the area of a triangle with base 15 cm and perpendicular height 8 cm.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A triangle has area 54 cm² and base 12 cm. Find its perpendicular height.

10. Write and solve a similar question to: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.

Teacher answers and checking prompts

1. Find the area of a triangle with base 15 cm and perpendicular height 8 cm.
Answer 60 cm².
2. A triangle has area 54 cm² and base 12 cm. Find its perpendicular height.
Answer 9 cm.
3. A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.
Answer 80 cm².
4. Solve this independently and show every stage: Find the area of a triangle with base 15 cm and perpendicular height 8 cm.
Answer 60 cm².
5. Use a second method or representation for: A triangle has area 54 cm² and base 12 cm. Find its perpendicular height.
Answer Expected result: 9 cm. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.
Answer Expected result: 80 cm². A valid check must be shown.
7. Prove or explain your reasoning: A rectangle is 14 cm by 9 cm. Find its area and perimeter.
Answer Area 126 cm²; perimeter 46 cm. The explanation must show why the method works.
8. Identify and correct this misconception: Miro uses the sloping side as the height and forgets to halve a triangle's area.
Answer Mark the perpendicular height, use base times height, and divide by two only for a triangle.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

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Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: A rectangle is 14 cm by 9 cm. Find its area and perimeter.

6. Create a new example that tests the same idea as: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.

2. Prove the answer to this question, rather than only calculating it: Find the area of a triangle with base 15 cm and perpendicular height 8 cm.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A triangle has area 54 cm² and base 12 cm. Find its perpendicular height.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 80 cm².

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro uses the sloping side as the height and forgets to halve a triangle's area.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A rectangle is 14 cm by 9 cm. Find its area and perimeter.
Answer Area 126 cm²; perimeter 46 cm. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Find the area of a triangle with base 15 cm and perpendicular height 8 cm.
Answer Expected result: 60 cm². Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A triangle has area 54 cm² and base 12 cm. Find its perpendicular height.
Answer Expected result: 9 cm. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 80 cm².
Answer One valid reconstruction is: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro uses the sloping side as the height and forgets to halve a triangle's area.
Answer Mark the perpendicular height, use base times height, and divide by two only for a triangle.
6. Create a new example that tests the same idea as: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.